

Temporal dependence - migraine

Patients 27; gap transitions 208 (max 37 days); missing-day fraction 0.0955.

Markov: $P(\text{attack tomorrow} \mid \text{attack today}) = 0.315$ vs 0.050 without (risk ratio 6.3). Self-excitation lag-1 OR 5.43 (trigger-controlled).

Pre-registered primary tests (raw p)

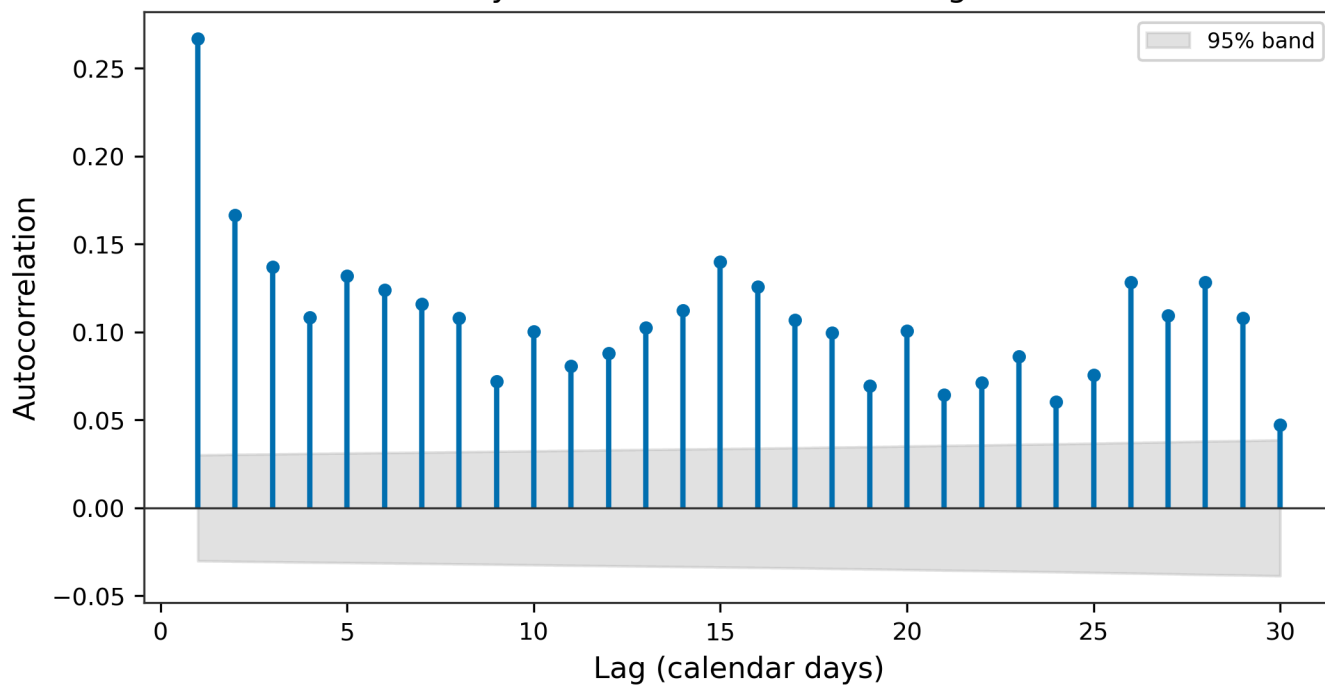
test	p
lag-1 Markov transition	8.76e-67
self-excitation LR (vs triggers-only)	2.48e-33
day-of-week omnibus	5.44e-01

Exploratory ACF lags (BH-FDR adjusted)

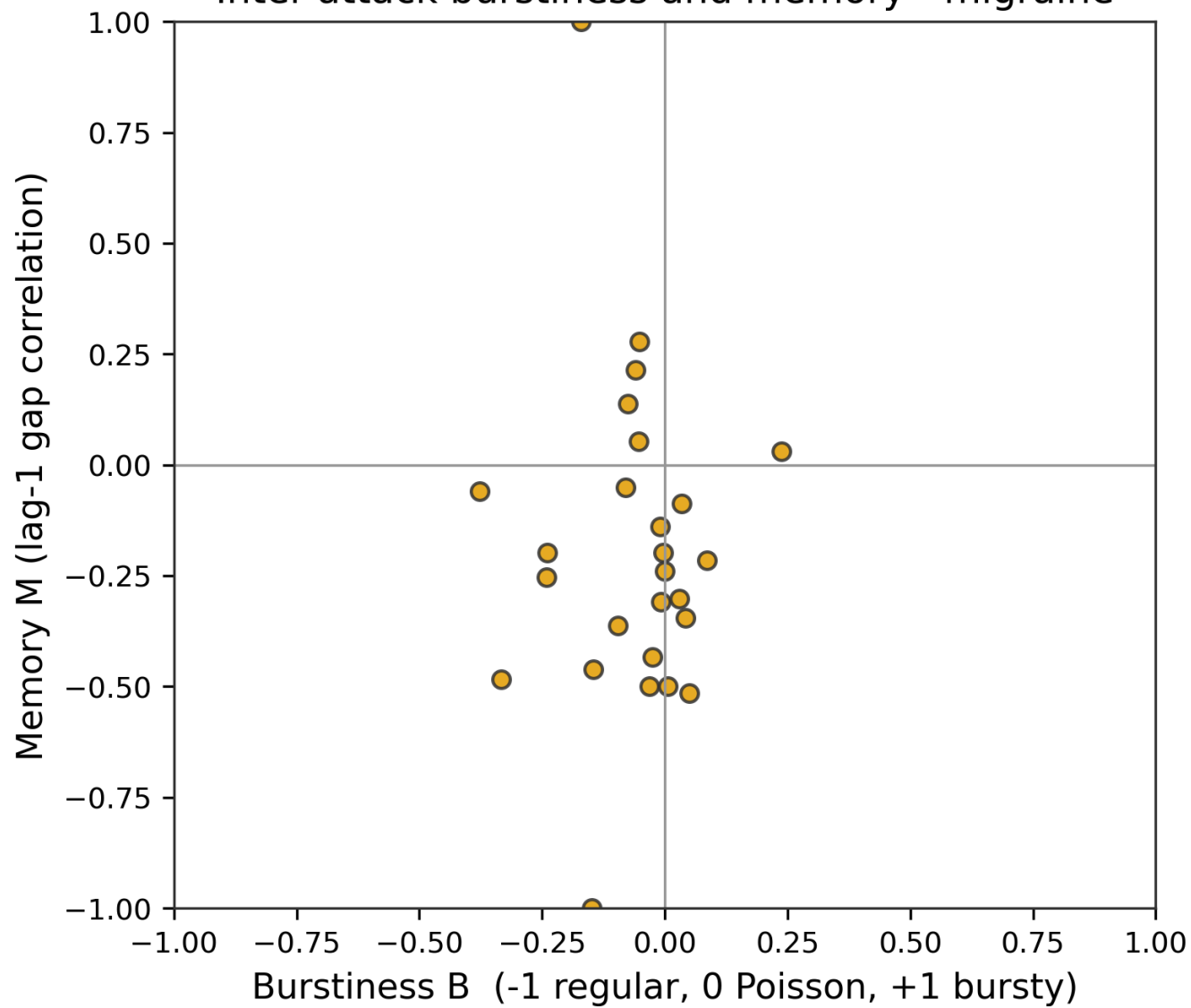
lag	p	p_adj
ACF lag 2	6.26e-27	8.14e-26
ACF lag 3	1.90e-18	1.24e-17
ACF lag 4	6.09e-12	1.32e-11
ACF lag 5	9.02e-17	3.91e-16
ACF lag 6	9.12e-15	2.97e-14
ACF lag 7	6.84e-13	1.78e-12
ACF lag 8	3.24e-11	6.02e-11
ACF lag 9	1.07e-05	1.07e-05
ACF lag 10	1.25e-09	1.79e-09
ACF lag 11	1.34e-06	1.45e-06
ACF lag 12	1.57e-07	1.85e-07
ACF lag 13	1.38e-09	1.79e-09
ACF lag 14	4.53e-11	7.35e-11

Figures

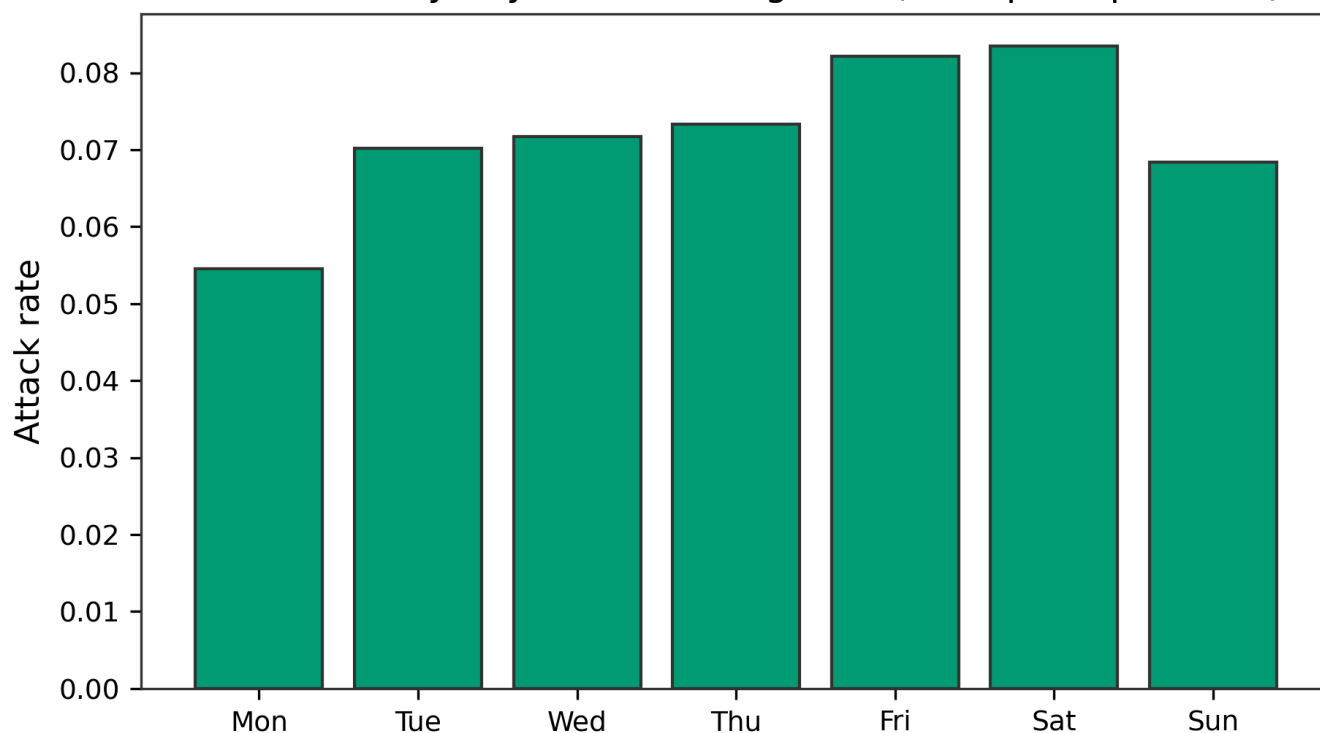
Daily attack autocorrelation - migraine



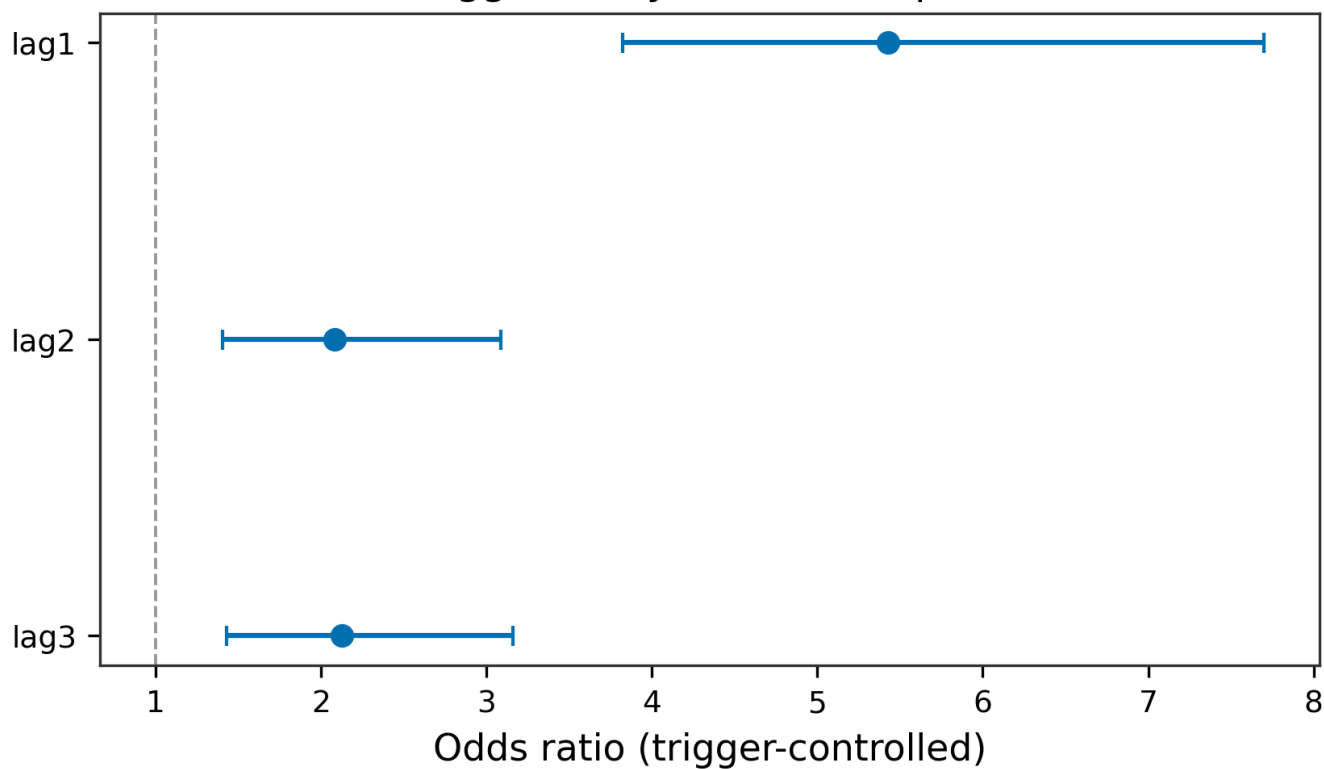
Inter-attack burstiness and memory - migraine



Attack rate by day of week - migraine (chi-square $p = 0.54$)



Attack-lag self-excitation - migraine
LR vs triggers-only: $\chi^2=155$, $p=2.5e-33$



First-order transition - migraine (RR=6.3, p=8.8e-67)

